



Original Article

Decoding whole genome of *Anoxybacillus rupiensis* TPH1 isolated from tatapani hot spring, India and giving insight into bioremediation ability of TPH1 via heavy metals and azo dyes

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ABSTRACT

A moderately thermophilic, gram-positive genomospecies *Anoxybacillus rupiensis* TPH1 was isolated from Tatapani hot spring, Chhattisgarh, India. Genome of 3.70 Mb with 42.3% GC subsumed 4131 CDSs, 65 tRNA, 5 rRNA, 35 AMR and 19 drug target genes. Further, comparative genomics of 19 *Anoxybacillus* spp. exhibited an open pan genome of 13102 genes along with core (10.62%), unique (43.5%) and accessory (45.9%) genes. Moreover, phylogenomic tree displayed clustering of *Anoxybacillus* spp. into two distinct clades where clade A species harbored larger genomes, more unique genes, CDS and hypothetical proteins than clade B species. Further, distribution of azoreductases showed FMN-binding NADPH azoreductase (AzoRed1) presence in clade A species only and FMN-binding NADH azoreductase (AzoRed2) harboring by species of both clades. Heavy metal resistance genes distribution showed omnipresence of *znuA*, *copZ* and *arsC* in both clades, dispersed presence of *cbiM*, *czcD*, *merA* and *feoB* over both clades and harboring of *nikA* and *acr3* by few species of clade A only. Additionally, molecular docking of AzoRed1, AzoRed2, ZnuA, CopZ, Acr3, CbiM, CzcD, MerA and NikA with their respective ligands indicated high affinity and stable binding. Conclusively, present study provided insight into gene repertoire of genus *Anoxybacillus* and a basis for the potential application of this thermophile in bioremediation of azo dyes and heavy metals.

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1. Introduction

Terrestrial hot springs are distinguished by emanation of meteoric water mixed with geothermal water, having temperature above 45 °C, from the fractures or faults formed on Earth's surface. Hot springs are research hot spots not only for the geologists and ecologists but also for microbiologists as they are considered as rich repertoire of thermophiles [1]. Hot springs are geographically-restricted special ecosystem and thus, offer a stable physicochemical environment to their endemic microbial community [2]. Hot springs are enriched with bacterial genera like *Chloroflexus*, *Acidithiobacillus*, *Sulfolobus*, *Thermus* etc. [3]. Among them, *Anoxybacillus* is one of the most common occupants of hot springs throughout the world.

The members of the genus *Anoxybacillus* (type species *Anoxybacillus pushchinonensis*) are gram-positive in nature, moderately thermophilic (having optimum growth temperature around 50–60 °C) and possess endospore producing ability. The cells are generally rod-shaped, straight or slightly curved, often paired or form short chains and have ability to survive at high pH. The genus *Anoxybacillus* was described by Pikuta et al. [4] on the basis of phenotypic characterization, 16S rRNA gene based phylogenetic analysis and DNA–DNA hybridization. After this description, the bacterium earlier described as *Bacillus flavothermus* was further reclassified to *A. flavithermus* as it was found to be a close relative of *A. pushchinonensis*. Later on, two species of *Geobacillus* – *Geobacillus tepidamans* and *Geobacillus caldoproteolyticus* were also reclassified to the genus *Anoxybacillus* using the polyphasic approach [5]. Presently, the genus *Anoxybacillus* consists of 25 species (24 validly described) and 2 subspecies (1 validly described) among which 14 are reported to be facultative anaerobes, 8 aerobes, 4 facultative aerobes and 1 aerotolerant anaerobe. Furthermore, only 19 members of this genus i.e., 18 species and 1

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Abbreviations:

AAI	Average amino acid identity
ANI	Average nucleotide identity
BPGA	Bacterial Pan Genome Analysis
CDS	Coding DNA sequence
COG	Cluster of Orthologous Groups
dDDH	digital DNA–DNA hybridization
GO	Gene Ontology
KEGG	Kyoto Encyclopedia of Genes and Genomes
CDART	Conserved Domain Architecture Retrieval Tool

subspecies, have been whole genome sequenced till now and only few reports are available where comparative genome analysis was accomplished to understand the diversity and distribution of gene pool among the genus *Anoxybacillus*.

Owing to the thermophilic nature, bioprospecting of genus *Anoxybacillus* was traversed and, prior experimental and genomic studies successfully showed industrially valuable enzymes harbored by *Anoxybacillus* spp. like amylases, proteases, cellulases, xylanases, lipases, carboxylesterases etc. and thus, conferring their industrial utilities [6]. Another bioprospecting aspect i.e., bioremediation potential of genus *Anoxybacillus* was examined where the presence of such genes and their function were documented. It was reported that few members like *A. contaminans*, *A. rupiensis*, *A. flavithermus*, *A. amylolyticus*, *A. gonensis* have significant efficiency to biosorb heavy metals like Hg, Cr, Cd etc. under laboratory conditions [6–8]. Further, capability of *Anoxybacillus* spp. in treating waste gases and thus, eventually reducing air pollutants, by acting as thermophilic biofilter has been experimentally proven [9]. Other than heavy metal detoxification and waste gases treatment, many species like *A. pushchinonensis*, *A. kamchatkensis*, *A. flavithermus*, *Anoxybacillus* sp. PDR2 have been demonstrated to degrade azo dyes (widely used in industries like food, textiles, cosmetics, pharmaceuticals etc.) and role of different enzymes in dye degradation like FMN-dependent NADH- azoreductase, NAD(P)H: quinone oxidoreductase and Catechol-2, 3-dioxygenase has been investigated [10]. Thus, the establishment of *Anoxybacillus* genus as an efficient degrader of potentially carcinogenic and mutagenic azo dyes was achieved.

Therefore, it is evident from the genomic and experimental studies that members belonging to this genus are not only the prolific sources of industrially valuable enzymes but also have considerable potential to be used as bioremediation agents for multiple types of pollutants. Thus, in the present study, a thermophilic strain of *Anoxybacillus* i.e., TPH1 was isolated from relatively unexplored Tatapani hot spring, Chhattisgarh, India and characterized by a combination of phenotypic, biochemical, 16S rRNA gene homology and phylogenomic analysis. Further, comparative genome analysis of the genus *Anoxybacillus* was performed to understand the proportion of genes involved in various cellular processes, with special emphasis on the distributional, structural and functional aspects of heavy metal detoxifying enzymes and azo dye degrading azoreductases.

2. Materials and methods

2.1. Sample collection and water analysis

Water sample was collected from Tatapani geothermal spring, covering an area of 0.1 km² in Balrampur district of Chhattisgarh, India (23°69'86" N, 83°68'40" E) in a sterile thermosteel vacuum

insulated bottles on 26th May, 2019 and subsequently, water analysis was performed [11].

2.2. Isolation & purification of thermophilic bacterial strain

The water sample collected from hot spring was serially diluted in sterile distilled water and inoculated on nutrient agar medium (peptone, 5 g/L; NaCl, 5 g/L; beef extract, 3 g/L) and incubated at 55 °C for 24 h. Morphologically distinct colonies were purified, individually sub-cultured and each colony was considered as unique strain. The strain, TPH1, exhibited higher growth rate at 55 °C for 24 h, thus, further studied.

The thermophilicity of purified TPH1 was assessed by observing its growth pattern at a temperature gradient (35 °C, 40 °C, 45 °C, 50 °C, 55 °C, 60 °C and 65 °C) in nutrient broth and the cell density was determined after 24 h of incubation at A₆₀₀. After qualifying, the strain TPH1 was conferred as thermophilic and was further characterized.

2.3. Colony morphology, physiology and biochemical characterization of TPH1

The attributes of pure TPH1 colony i.e., the shape, margin, color, elevation, appearance, opacity and texture, was studied using light microscope (Dewinter Optical Inc. India).

For growth analysis, TPH1 was inoculated (A₆₀₀ = 0.1/mL) into 100 mL nutrient broth and incubated at 55 °C. The absorbance was periodically recorded at 600 nm till the stationary phase was achieved. Further, the growth of TPH1 was also assessed at varying pH (5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5 and 10.0) and NaCl concentrations (0.5%, 1%, 1.5%, 2%, 2.5%, 3%, 3.5%, 4.0%, 4.5% and 5%). In this case, the cell growth was estimated after 7 h by measuring the absorbance at 600 nm. All the experiment were performed in triplicates.

Furthermore, the biochemical attributes such as gram staining, motility, citrate utilization ability, urease activity, Methyl Red - Vogue Proskauer (MR-VP) test, casein hydrolysis and carbohydrate fermentation test (using glucose, dextrose and fructose) was performed as described in Harrigan and McCance [12]. The nitrate reducing ability was tested as mentioned in Mishra et al. [13]. Further, superoxide dismutase (SOD) [14] and catalase (CAT) [15] assays were performed to test the enzymatic antioxidative attributes of TPH1. Amylolytic and cellulolytic potential of the TPH1 was studied by observing formation of zone of hydrolysis on starch agar and carboxymethyl cellulose agar (CMC) plate, respectively [16]. Also, the gelatin hydrolysis was checked using the method described in Mohammad et al. [16]. Substrate utilizing ability and catabolic potential of TPH1 for 94 different carbon sources was assessed using Biolog (GEN III) plate as per manufacturer's protocol. Upon adding requisite inoculum to each well, the plate was incubated at 55 °C for 7 h and the absorbance was recorded at 590 nm and 720 nm. Cellular fatty acids profiling was performed by GC–MS analysis of the prepared fatty acid methyl esters (FAMES) using the protocol described by El Razak et al. [17].

2.4. Genomic DNA extraction and 16S rRNA gene sequencing

The genomic DNA of TPH1 was extracted by phenol/chloroform method of Sambrook et al. [18]. Further, the 16S rRNA gene was amplified using universal primers 27F [5'-CCA GAG TTT GAT CMT GGC TCA G-3'] and 1492R [5'-TAC GGY TAC CTT GTT ACG ACT T-3'], and the amplicon was purified by standard polyethylene glycol (PEG)-NaCl precipitation. This purified product was sequenced using Sanger's method on an ABI® 3730XL automated DNA sequencer (Applied Biosystems, Inc., Foster City, CA) as per

manufacturer's protocol. For identification, the 16S rRNA sequence of TPH1 was compared against EzBioCloud database (<https://www.ezbiocloud.net/>) [19].

2.5. Whole-genome sequencing, assembly and annotation

40 ng of genomic DNA was amplified using 10 pM of corresponding forward and reverse primers. The amplicons were purified with Ampure beads and eight additional PCR cycles were performed using Illumina barcoded adapters to prepare the sequencing libraries. Further, the libraries were also purified using Ampure beads and quantified using Qubit dsDNA High Sensitivity assay kit. Finally, the sequencing was performed using Illumina Miseq.

PATRIC 3.6.9 services were (<https://www.patricbrc.org/>) used for assembly and annotation of raw reads (FastQ). Comprehensive Genome Analysis Service assembled the sequences by Unicycler v4.0, using genome of *A. rupiensis* DSM 17127 (1297581) as reference, based on 16S rRNA phylogeny. Further, RAST tool kit (RASTtk) based Genome Annotation Service of PATRIC 3.6.9 was used for annotation and assigning subsystems to the genome of TPH1. Additionally, the functional Gene Ontology (GO) categorization of annotated proteins and distribution of enzyme classes in TPH1 genome was analyzed using OmicsBox. General genomic features like length, contigs, tRNA, rRNA counts, GC content (%), coding sequences (CDS), anti-microbial resistance genes (AMR), virulence factor (VF), transporters and drug target genes, etc. were also retrieved from PATRIC 3.6.9.

2.6. Genome similarity measures

Genome similarity of TPH1 with *Anoxybacillus* spp. was determined in terms of average nucleotide identity (ANI), average amino acid identity (AAI) and digital DNA–DNA hybridization (dDDH). ANI was calculated using OrthoANLu (<https://www.ezbiocloud.net/>) and ANIb (<http://jspecies.ribohost.com/jspeciesws/>) and the cut off value of 95% was used for species-delineation. The AAI was calculated by Kostas lab server (<http://enve-omics.ce.gatech.edu/>) and result from Two-Way AAI was adopted for species-delineation at a cut off value of 95%. Besides, dDDH was performed using the Genome-to-Genome Distance Calculator (GGDC) (<http://ggdc.dsmz.de/home.php>).

2.7. Construction of phylogenetic trees and analysis

The 16S rRNA gene-based phylogenetic tree of *Anoxybacillus* spp. was constructed by Neighbor-joining method [20] using MEGA v7.0 software [21]. This involved 28 nucleotide sequences including *Bacillus subtilis* subsp. *subtilis* DSM 10 (ABQ101000001) as outgroup. The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances, which were computed using the Tamura 3-parameter method [22]. The reliability of each branch was corroborated by 1000 bootstrap replications and the bootstrap values above 50 were shown next to branches.

At the time of analysis, whole genome sequences of 18 *Anoxybacillus* spp. were available and thus, the codon tree was constructed using 19 genomes of *Anoxybacillus* spp. including TPH1 by comparing 1000 single-copy genes, having *Bacillus subtilis* subsp. *subtilis* DSM 10 as the outgroup, using phylogenetic tree building service of PATRIC 3.6.9. A concatenated alignment was generated for all proteins and coding DNA sequences which was automatically redirected towards RAXML for supertree construction by maximum-likelihood with 1000 bootstrap replication. The tree was further visualized using iTOL server (Interactive Tree Of Life) (<https://itol.embl.de/login.cgi>) [23].

2.8. Pan/core-genome, Cluster of Orthologous Groups (COGs) and Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis

The 19 genomes of *Anoxybacillus* spp. were subjected to pan genome analysis. The core (conserved), accessory (dispensable), and unique (strain-specific) genes were analyzed using the Bacterial Pan Genome Analysis (BPGA) pipeline with default settings [24]. Further, BPGA was used to identify the Cluster of Orthologous Groups (COGs) among these *Anoxybacillus* spp. Additionally, the Kyoto Encyclopedia of Genes and Genomes (KEGG) database was accessed through BPGA for understanding the functional attributes of genes among these 19 genomes. The gene families associated with “metabolism”, “genetic information processing”, “environmental information processing”, and “cellular processes” etc. were examined.

2.9. Distribution of genes associated with heavy metal resistance and azo dye degradation

The genes conferring resistance ability against heavy metals i.e., Zn, Cu, Fe Co, Ni, Cd, Hg, As, and Cr in TPH1 were explored. For this, genes that are associated with export and import of heavy metals, metalloenzymes, metallochaperones and metal-sensing transcriptional regulators were enlisted and the distribution of selected genes related to each metal homeostasis among the *Anoxybacillus* genus was studied. The domain identification and analysis of all the retrieved protein sequences corresponding to each selected gene was performed using NCBI Conserved Domain Architecture Retrieval Tool (CDART) [25].

Further, the gene expressing azo-dye degrading enzyme i.e., azoreductase was selected for study and protein sequences corresponding to azoreductase were investigated in all the genomes of *Anoxybacillus* to confer its distribution and abundance. Further, the domain identification and analysis of all the retrieved sequences was performed using CDART [25].

2.10. Theoretical modeling of metal homeostasis related enzymes and azo-dye degrading enzymes

High quality homology models of znuA (fig|311460.11.peg.26), copZ (fig|311460.11.peg.2031), czcD (fig|311460.11.peg.602), feoB (fig|311460.11.peg.1323), nikA (fig|311460.11.peg.3683), cbiM (fig|311460.11.peg.2683), merA (fig|311460.11.peg.2036), arsC (fig|311460.12.peg.2561), acr3 (fig|311460.12.peg.2560), AzoRed1 (fig|311460.12.peg.2389) and AzoRed2 (fig|311460.12.peg.3032) were generated using Swiss-Model (<https://swissmodel.expasy.org/interactive>) [26–30]. Finally, the quality of homology models was validated by Ramachandran plot using PROCHECK (<https://servicesn.mbi.ucla.edu/PROCHECK/>) [31].

2.11. Molecular docking of metal homeostasis related enzymes and azo-dye degrading azoreductase enzyme

The ability of certain enzymes in providing heavy metal tolerance and azo-dye degradation to TPH1 was examined by computational docking. The ligands used were Zn²⁺, Cu²⁺, As⁵⁺, Hg²⁺, Co²⁺, Ni²⁺, Cd²⁺ and an azo dye MO (methyl orange). *In silico* docking of ligands with their respective enzymes was performed using Autodock vina [32]. Firstly, each of the protein and ligands were prepared using Autodock tools and saved in pdbqt format. Subsequently, grid box was defined on the protein molecules to enable docking. Eventually, docking was performed through vina.exe. The docking poses with lowest binding energies were visualized in PyMOL2 [33] and UCSF Chimera v.1.1.3 [34]. Metal- and methyl orange-interacting amino acid residues of respective

proteins were visualized using LigPlot+ and Biovia Discovery Studio 2021 Visualizer.

2.12. Determination of growth responses of TPH1 against heavy metals and an azo dye methyl orange (MO)

TPH1 was inoculated ($A_{600} = 1.0/\text{mL}$) into 100 mL of nutrient broth having varying concentrations of heavy metals i.e., ZnSO_4 for Zn^{2+} (0.05, 0.1, 0.2 and 0.3 mM), CuSO_4 for Cu^{2+} (0.25, 0.5 and 0.1 mM), FeCl_3 for Fe^{3+} (0.2, 0.4, 0.6, 0.8 and 1 mM), FeSO_4 for Fe^{2+} (0.1, 0.25, 0.5 and 1 mM), NaAsO_2 for As^{3+} (0.5, 1 and 2 mM), Na_3AsO_4 for As^{5+} (1, 2, 5 and 6 mM), HgCl_2 for Hg^{2+} (0.01, 0.025 and 0.05 mM), CoCl_2 for Co^{2+} (0.1, 0.2, 0.3 and 0.4 mM), NiSO_4 for Ni^{2+} (0.4, 0.8, 1.2 and 1.6 mM), CdCl_2 for Cd^{2+} (0.25, 0.5, 1 and 5 mM) and $\text{K}_2\text{Cr}_2\text{O}_7$ for Cr^{3+} (0.05, 0.1 and 0.2 mM). After 24 h of incubation, absorbance was recorded at 600 nm and consequently, LC_{50} was determined by conducting probit analysis.

Further, TPH1 was inoculated ($A_{600} = 1/\text{mL}$) into 100 mL of nutrient broth having different concentrations of an azo dye i.e., MO (50 mg/L, 100 mg/L, 200 mg/L, 300 mg/L and 400 mg/L) and incubated at 55 °C. At every 2 h of interval, 1 ml of culture was harvested and A_{600} nm was recorded. At exponential phase, LC_{50} was determined and decolorization % was calculated [35].

3. Results

3.1. Hot spring water analysis

The hot spring had a temperature of 85 °C, pH 8.8, salinity 310 ppm, total dissolved solids (TDS) 722 ppm and conductivity 720 μS . The water was rich in Na^+ i.e., 822 mg/L and had sulfur content of 87 mg/L (Table S1).

3.2. Colony morphology and physiology of TPH1

The colony of TPH1 was circular shaped, creamy white, shiny, smooth and opaque with entire margin, flat elevation and dry texture (Table S2). The cells of TPH1 appeared rod-shaped. Further, the optimum growth temperature of TPH1 was found to be 55 °C and the generation time was 2.19 h. The growth under different pH regimes displayed maximum growth at pH 6.0, whereas no growth was observed below pH 5.5 and above pH 9.0 (Table S2). Additionally, the bacterium showed optimal growth at 1% NaCl concentration whereas 0.5% and 2% NaCl concentration were sub-optimal and concentration above 2% was found to be lethal (Table S2).

3.3. Biochemical attributes and substrate utilizing ability of TPH1

The cells of TPH1 were motile, gram-positive and showed negative test for citrate utilization, urease and MR-VP, whereas positive result was attained for nitrate reduction. The antioxidative activity of bacterium was conferred by positive test for SOD and CAT. Further, TPH1 formed a clear zone of hydrolysis on starch agar and skimmed milk agar plate which indicated a positive result for amylase and caseinase, respectively. However negative results were observed for cellulase activity as no clear transparent zone was formed on CMC agar plate and for gelatinase enzyme activity (Table S2). The major cellular fatty acids in TPH1 were found to be iso-C_{15:0} (53%) and iso-C_{17:0} (24%) (Table S3).

Among various carbohydrate substrates, TPH1 utilized D-Trehalose, Stachyose, D-Turanose, D-Raffinose, D-Galactose, α -D-Glucose, D-Mannose, 3-methyl glucose, Inosine, D-Glucose-6-phosphate and D-Fructose-6-phosphate. Among amino acids, L-Alanine, L-Histidine, L-Glutamic acid, L-Arginine, Glycyl-L-Proline, D-Glutamic acid, D-Aspartic acid, L-Aspartic acid and L-Pyroglutamic Acid showed to

favor bacterial growth. Hexose acids/sugar acids like D-Galactouronic Acid, L-Galactonic Acid Lactone, D-Gluconic Acid, D-Glucouronic Acid, Glucouronamide, Mucic Acid and Quinic Acid were also utilized by TPH1. Moreover, sugar alcohols such as D-Mannitol, D-Arabitol, myo-Inositol and Glycerol were readily utilized. Besides, only N-Acetyl Neuraminic Acid was utilized by TPH1 among the amino sugars. Among the mixed sources like carboxylic acids, esters and fatty acids, D-Saccharic Acid, Citric Acid, α -hydroxy-Butyric Acid, β -hydroxy-D, L-Butyric acid, D-Lactic Acid Methyl Ester, L-Lactic Acid, D-Malic acid and L-Malic acid were utilized. Comparing the rate of utilization of individual carbon compounds, it was observed that D-Trehalose, Inosine, D-mannitol, D-Fructose-6-phosphate, Glycerol, L-Histidine, L-Glutamic acid, D-Galactouronic Acid, L-Galactonic Acid Lactone, D-Gluconic Acid, D-Glucouronic Acid, L-Malic acid and D-Glucouronamide were intensively utilized whereas substrates like D-Fructose, D-Turanose, D-Raffinose, D-Galactose, D-Mannose, L-Arginine etc. were catabolized at a very low rate. Besides, few substrates like stachyose, D-Glucose-6-phosphate, L-Pyroglutamic Acid, Mucic Acid, myo-Inositol, L-Alanine etc. were utilized at relatively higher rate than former (Table S4).

3.4. 16S rRNA gene phylogenetic analysis

The 16S rRNA gene percentage similarity of TPH1 was found to be 99.7% with *A. rubeus* DSM 17127 and *A. geothermophilus* ATCC BAA-2555. Further, in the phylogenetic tree reconstructed using the 16S rRNA gene sequences of 27 *Anoxybacillus* spp., TPH1 formed a strong cluster with *A. rubeus* DSM 17127 and *A. geothermophilus* ATCC BAA-2555 (Fig. 1A).

3.5. Phylogenomic analysis

The phylogenomic (codon) tree constructed using 19 genomes of *Anoxybacillus* spp. exhibited that TPH1 formed strong cluster (100 bootstraps) with *A. rubeus* DSM 17127, whereas *A. geothermophilus* ATCC BAA-2555 shared relatively distant relationship. Further, the comparison of genomic features like OrthoANIu, ANIb, dDDH and AAI showed that the TPH1 possessed 99.43%, 99.2%, 94.7% and 100% similarity with *A. rubeus* DSM 17127, whereas only 98.64%, 98.3%, 86% and 97.77% similarity with *A. geothermophilus* ATCC BAA-2555, respectively (Fig. 1B).

3.6. General genomic features of TPH1

The genomic size of TPH1 was 3709752 bp with 96.8% completeness and the total number of contigs were found to be 22. The coarse consistency and fine consistency were found to be 98.9% and 97.3%, respectively. Further, the N50 and L50 values were 436469 bp and 3, respectively, while the GC content was 42.3% (Table S5). The genome subsumed 65 tRNA, 5 rRNA, 4131 coding sequences (CDS), 258 CRISPR repeats and 250 CRISPR spacers. Specialty genes identified in the TPH1 genome consisted of 35 anti-microbial resistance genes (AMR), 6 virulence factor (VF), 12 transporters and 19 drug target genes (Fig. 2A). The genome of TPH1 possess genes distributed over 26 subsystems in which amino acid & its derivatives subsystem has highest no. of genes i.e., 319 whereas carbohydrates subsystem has second highest no. of genes i.e., 214. Potassium metabolism and cell division & cycle subsystem have least no. of genes related to it i.e., 2 & 4 respectively (Table S6).

3.7. Gene ontology-based categorization of annotated proteins and distribution of enzyme classes

Annotated proteins of TPH1 were broadly categorized either into biological or molecular functions based on the gene ontology.

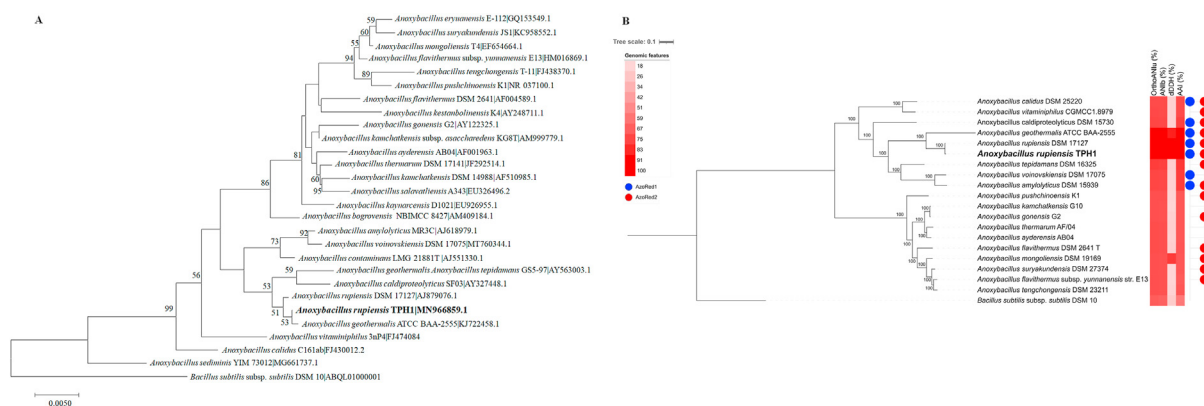


Fig. 1. Phylogenetic relationship among the *Anoxybacillus* spp (A) Rooted tree based on 16S rRNA gene sequence of 27 *Anoxybacillus* spp. was constructed in the neighbor-joining method with *Bacillus subtilis* subsp. *subtilis* DSM 10 (ABQ01000001) as outgroup. The optimal tree has the sum of branch length = 0.16468766. The evolutionary distances were computed using the Tamura 3-parameter method, which are in the units of the number of transitional substitutions per site. Bar display 0.005 nucleotide substitutions per site. Branch reliability was corroborated by 1000 bootstrap replications. Bootstrap values above 50 were shown next to branches (B) The phylogenomic tree of 19 *Anoxybacillus ruiensis* TPH1 was constructed with *Bacillus subtilis* subsp. *subtilis* DSM 10 as the outgroup. The tree was based on aligned concatenated sequences of 1000 single-copy gene families using maximum-likelihood. The bootstrap values indicated before each node represented high degree of confidence for cluster formation.

In biological function, maximum proteins i.e., 1672, were associated with the cellular process (GO:0009987), followed by metabolic process (GO:0008152) consisting 1419 proteins. The total genes responsible for localization (GO:0051179), biological regulation (GO:0065007), response to stimulus (GO:0050896), signaling (GO:0023052) and developmental processes (GO:0032502) were 355, 308, 222, 88 and 78, respectively. Besides, other proteins were involved in multi-organism processes, positive and negative regulation of biological process and processes involved in intraspecies interaction, biological adhesion, locomotion and detoxification etc (Fig. 2B).

Most of the proteins involved in molecular functions were associated with catalytic activity (GO:0003824; 1717 proteins) or binding (GO:0005488; 1228 proteins), whereas a smaller percent of proteins was associated with transcription regulator activity (GO:0140110; 92), structural molecule activity (GO:0005198; 60) and transporter activity (GO:0005215; 228). Other proteins were responsible for small molecule sensor activity, antioxidant activity, translation regulator activity, molecular transducer activity, molecular carrier activity, molecular function regulator, nutrient reservoir activity and molecular adaptor activity (Fig. 2B).

Distribution of enzyme classes was uneven in the genome of TPH1. It was observed that the hydrolases and transferases accounted for 29% and 28% of all the enzymes, whereas isomerases and ligases each subsumed 5% of the enzymes. Besides, 15%, 7% and 11% enzymes were oxidoreductases, lyases and translocases, respectively (Fig. 2C).

3.8. Analyzing the genome of TPH1 for the presence of genes expressing industrially valuable enzymes including exopolysaccharide biosynthesis enzymes

The annotated genome of TPH1 was found to possess many economically important enzymes like carboxylesterase, endo-1,4-beta-xylanase, xylosidase, α -glucosidase, 6-phospho- β -glucosidase, oligo-1,6-glucosidase, α -galactosidase, lipase/acylhydrolase, lipase/acylhydrolase with GDSL-like motif, lysophospholipase, glycosyl transferase, group 1 family protein, glycosyl transferase, group 2 family, glycosyltransferase 28-like, L-asparaginase, manganese catalase, cyanoalanine nitrilase, glutathione peroxidase, thiol peroxidase, neopullulanase, pullulanase, peptidase S8 and S53, subtilisin, kexin, sedolisin and transglutaminase-like (Table S7

A). Further, 46 genes related to exopolysaccharide biosynthesis were manually identified in the genome of TPH1 (Table S7 B).

3.9. Pan genome, orthologue clustering and KEGG analysis

Analyzing the genomes of 19 *Anoxybacillus* spp. revealed a pan genome of 13102 genes, of which 1392 genes, accounting for 10.62% of pan genome, represented the core genome. Also, the core genes occupied 38.89% of the average CDS among these 19 strains. Further, 29954 and 5695 genes represented accessory and unique genes, respectively. The maximum number of unique genes (26%) were found in *A. geothermophilus* ATCC BAA 2555, whereas no unique genes were harbored by *A. ruiensis* DSM 17127 and *A. ruiensis* TPH1. The pan genome curve of 19 *Anoxybacillus* spp. was asymptotic and defined by $P_s[n] = 3275.87 n^{0.470789}$ based on power law regression, where $y = 0.470789$ ranged between 0 and 1, indicated open pan genome. In contrast, the core genes were relatively constant and defined by exponential regression i.e., $F_c[n] = 2045.3e^{0.00282964n}$. The curve of core genome followed a gradual decline and reduced to 1194 genes upon incorporation of 19th genome (Fig. 3).

Functional analysis of COGs in 19 *Anoxybacillus* spp. showed that the majority of core, accessory and unique genes were involved in "metabolism" whereas, least genes were associated with "cellular processes and signaling". Additionally, "information storage and processing" accounted for 28%, 26% and 30% of core, accessory and unique genes, respectively (Fig. 4A). Further, the functional prediction of pan genome revealed "general function" [R] accounted for maximum number of COGs. On the other hand, "cell cycle control, cell division, chromosome partitioning" [D], "intracellular trafficking, secretion and vesicular transport" [U], and "nucleotide transport and metabolism" [F] have lesser shares of COGs (Fig. 4C).

Furthermore, the KEGG analysis in these genomes revealed that highest percentage of the core, accessory and unique genes were involved in "metabolism" whereas, lowest percentage was associated with "organismal systems". For "environmental information processing" higher percent of accessory genes were associated than core and unique genes, however, in "genetic information processing" more percentage of core genes were involved than accessory and unique genes (Fig. 4B). Further, it was observed that highest percentage of core, accessory and unique genes were associated with "carbohydrate metabolism". While, "amino acid metabolism"

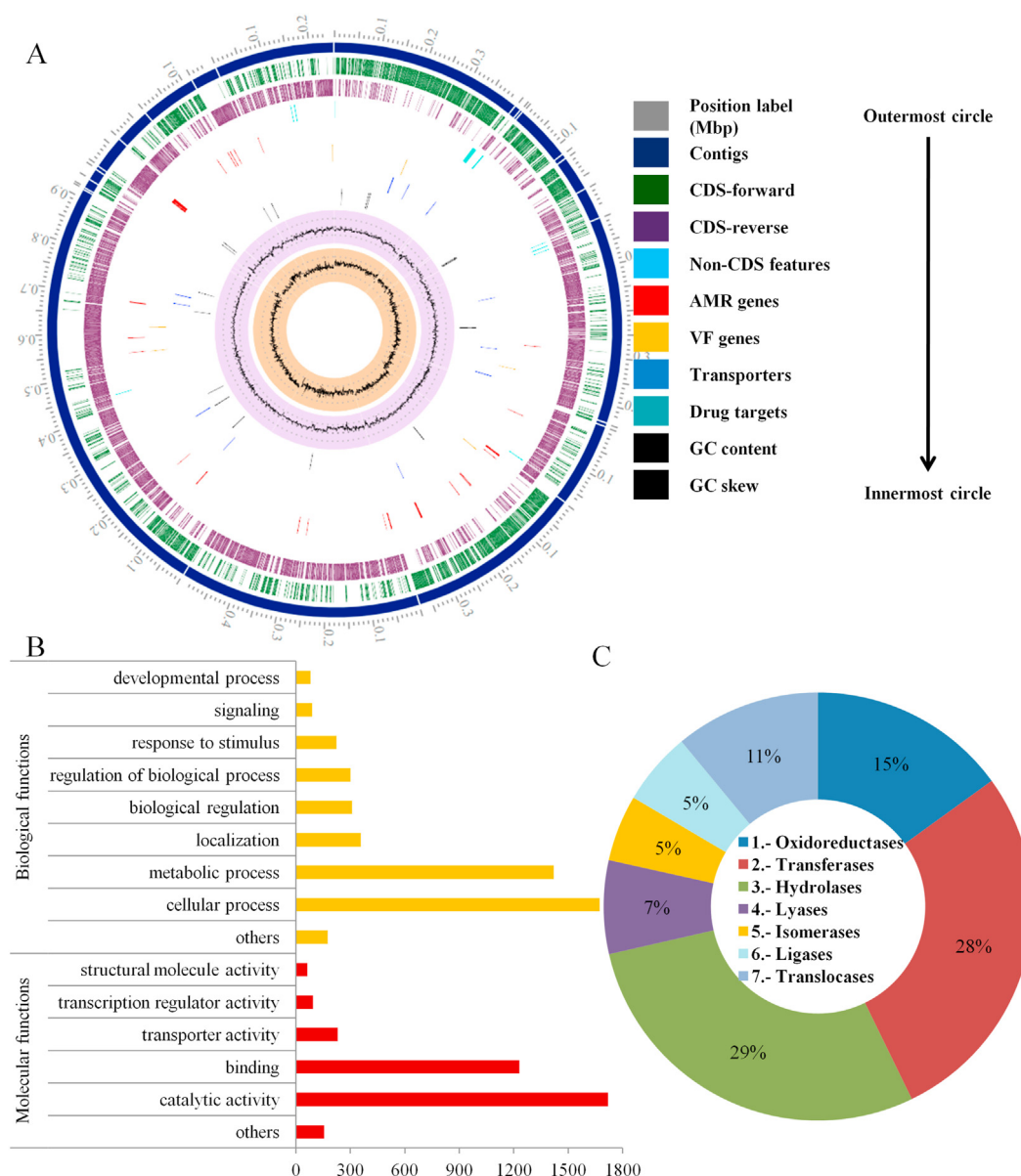


Fig. 2. (A) Circular plot showing genome characteristics of *Anoxybacillus rupiensis* TPH1. The information is displayed from the outer to innermost circle, as follows: position label (Mbp), contigs, CDSs on the forward strand, CDSs on the reverse strand, non-CDS features, AMR genes, VF genes, Transporters, drug targets, GC content and GC skew. CDSs, coding sequences; AMR, anti-microbial resistance; VF, virulence factors; GC, guanine-cytosine (B) Categorization of annotated proteins into biological and molecular functions on the basis of Gene Ontology (C) Distribution of enzyme classes in *Anoxybacillus rupiensis* TPH1.

also consisted of higher percentage of core and unique genes, more of the accessory genes were involved in “membrane transport”. Nevertheless, lowest percentage of genes was associated with the processes such as “signaling molecular and interactions”, “substance dependency”, “sensory system” and “development” (Fig. 4D).

3.10. Assessing the heavy metal resistance and azo dye degradation related genes in the genome of TPH1

The genes related to heavy metal resistance were investigated in the annotated genome of TPH1 and it was revealed that the bacterium possesses several transporters related to efflux and influx of heavy metals, metal-binding enzymes, metalloregulators and metallochaperones (Table S8 A). The distribution of the selected

genes within genus *Anoxybacillus* was uneven. Genes like *znuA*, *copZ* and *arsC* were present in all the genomes of *Anoxybacillus* genus whereas dispersed presence of *nika*, *czcD*, *cbiM*, *merA* and *acr3* genes was observed (Table S8 B).

The genome analysis of TPH1 revealed the presence of genes for two types of azo dye degrading azoreductases i.e., FMN-binding NADPH azoreductase (AzoRed1) and FMN-binding NADH azoreductase (AzoRed2). While both the azoreductase types consisted of FMN binding domain, the former utilizes NADPH as the electron donor whereas NADH is being used by the latter type for its activity (Table S9). However, distribution of these azoreductases was uneven among the genus *Anoxybacillus* since, FMN-binding NADH azoreductase was harbored by the members of both clade A and clade B, the FMN-binding NADPH azoreductase was distributed only within the members of the clade A (Fig. 1B).

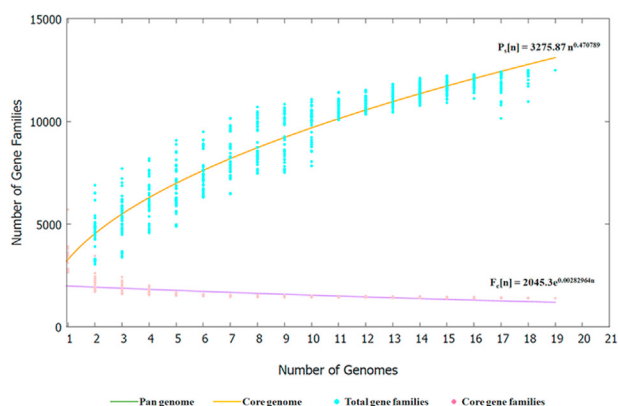


Fig. 3. Pan-genome and core genome of 19 *Anoxybacillus* spp. The number of genes in the pan and core are represented as cyan and pink color respectively. These genes are plotted as a function of the genomes added to the plot. The trends of pan and core genome are indicated by orange and purple lines curves. The equations fit to respective curves are shown next to them. While pan genome is open, the core genome gradually declined with addition of new genomes. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3.11. Comparative structural and docking analysis of azoreductases

Theoretical modeling of AzoRed1 and AzoRed2 with suitable templates displayed dimeric conformations for both the proteins (Fig. 5A and B). Fold analysis of these azoreductases displayed similar α/β topology (Fig. 5C and D). The bond angle stereochemistry analyzed using Ramachandran plot depicted >90% residues in the most favored region, conferring good quality of the homology models. Additionally, the root mean square deviations (RMSD) of

these models with their respective templates were <1 Å, further validated their high quality (Fig. S1). Additionally, the molecular docking of MO with AzoRed1 and AzoRed2 exhibited stable interactions due to lower binding energies of −7.2 kcal/mol and −7.7 kcal/mol, respectively. The MO binding pocket of AzoRed1 comprised of S9, R11, G14, K15, T16, P72, E73, Y74, V104, G105, G106, G110, and G124. The MO-AzoRed1 docked complex was stabilized by hydrogen bonding and van der Waal forces in association with T-shaped Pi–Pi and Pi-alkyl interactions. On the other hand, MO-AzoRed2 docked complex was stabilized by hydrogen bonding, van der Waal forces, Pi-sulfur, T-shaped Pi–Pi, Pi-alkyl, and alkyl–alkyl interactions. The binding pocket of AzoRed2 consisted of H10, S17, Y18, S19, M20, P101, F102, W103, N104, F105, Y127, P132, A146, S147, G148, G149, Y151, F172 and H186 (Fig. 6).

3.12. Evaluation of metal binding abilities of selected proteins

The docking of metals with respective proteins suggested stable interaction due to lower binding energies. The binding energy was −1.2 kcal/mol for CzcD_Cd, Hgred_Hg and NiTr_Ni docked complexes whereas a binding energy of −1.3 kcal/mol was exhibited by docked complexes of cbiM_Co, Acr3_As and ZnuA_Zn. The docked complex of CopZ_Cu had binding energy of −1.4 kcal/mol. The binding pocket of ZnuA (H71, H158, E297 and H222), CzcD (A142, L146, G145 S182 and I186), CbiM (S141, Q186, A137 and N138), CopZ (L26, D27, G28, and V29), MerA (G391, T394 and Q406), Acr3 (C124, V128, S296 and T293) and NikA (I475, T453, I449, A450 and F452) was depicted in Fig. 7. The van der Waals interaction was observed for NikA, CzcD, cbiM, CopZ, MerA and Acr3 with their respective ligands, whereas the interaction of ZnuA with Zn was stabilized by hydrogen bonds.

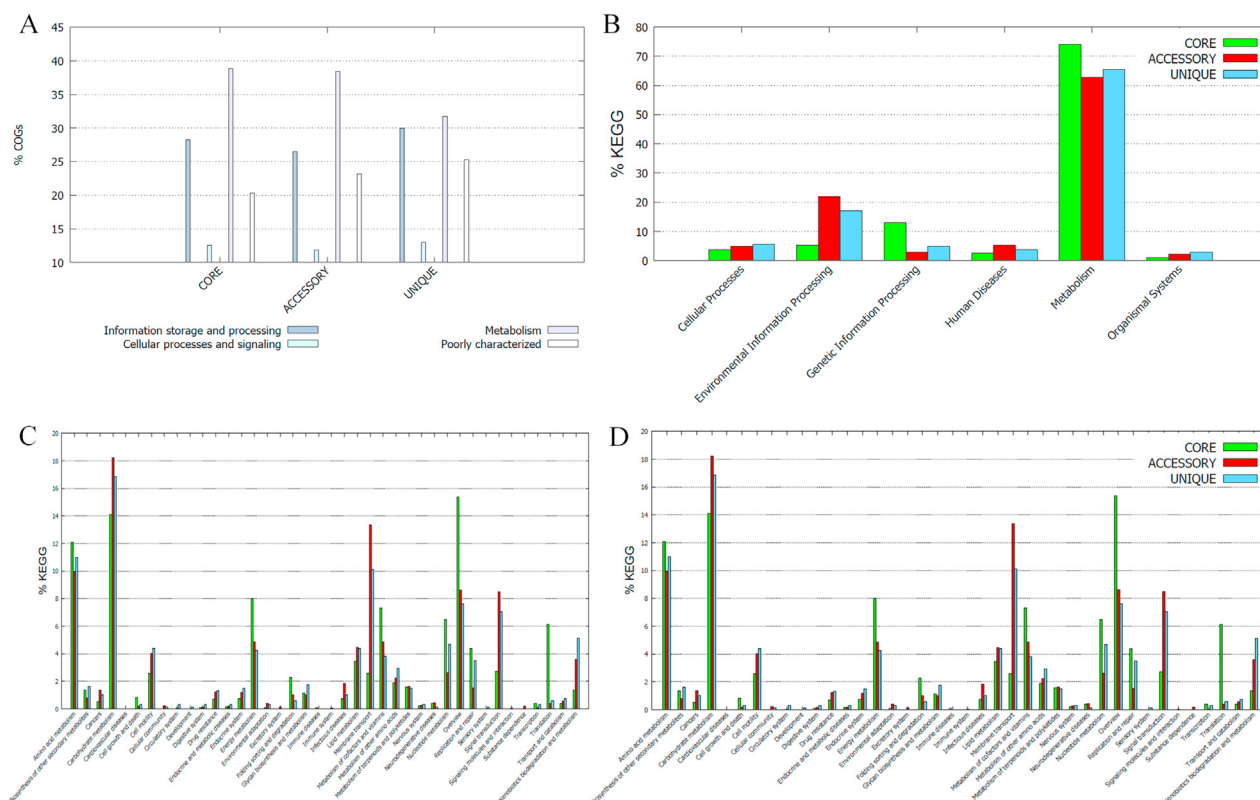


Fig. 4. COG and KEGG analysis of the pan genome in 19 *Anoxybacillus* strains (A) Distribution of functional COGs in each core, accessory, and unique genome (B) Percentage distribution of KEGG genes assigned in each core, accessory, and unique genome (C) Detailed distribution of COGs with their associated functions (D) Detailed distribution core, accessory, and unique genome in various pathways and metabolisms based on KEGG database.

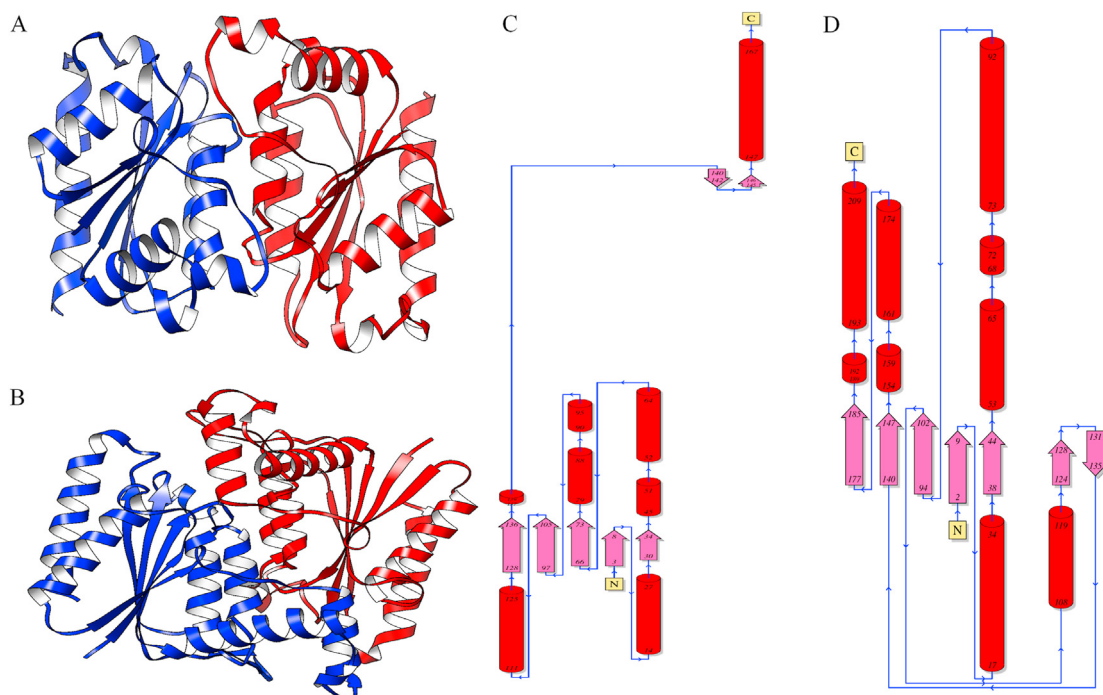


Fig. 5. Analysis of the homology models of azoreductases (A) & (C). The 3-D structure of AzoRed1 and Topology of secondary structure of AzoRed1 showing the arrangement of 7 β -sheets and 8 α -helices (B) & (D). The 3-D structure of AzoRed2 and Topology of secondary structure of AzoRed2 showing the arrangement of 7 β -sheets and 9 α -helices.

3.13. Growth responses of TPH1 against methyl orange and heavy metals

The growth curve of TPH1 in the presence of different concentrations of MO depicted that 100 mg/L of MO reduced ~50% of growth of TPH1 at exponential phase and hence, considered as LC_{50} (Fig. S3).

Probit analysis was performed for obtaining the LC_{50} for various metals which was found to be in the range of 0.029–5.79 mM. The highest growth inhibiting heavy metal for TPH1 was Hg^{2+} whereas As^{5+} was required to be in high concentration for similar growth inhibition. The descending order of LC_{50} for various metals is - As^{5+} (5.79 mM), Ni^{2+} (1.59 mM), Fe^{2+} (0.85 mM), Fe^{3+} (0.61 mM), Cd^{2+} (0.54 mM), As^{3+} (0.5 mM), Cu^{2+} (0.49 mM), Co^{2+} (0.22 mM), Zn^{2+} (0.144 mM), Cr^{3+} (0.122 mM) and Hg^{2+} (0.029 mM) (Fig. S4).

4. Discussion

A bacterium TPH1 was isolated from the Tatapani geothermal spring located in the Balrampur district of Chhattisgarh, India with OGT at 55 °C indicating its moderate thermophilic nature. Initial attempt to identify TPH1 using the 16S rDNA displayed 99.7% similarity with *A. rupiensis* DSM 17127 and *A. geothermalis* ATCC BAA-2555. Also, the reconstructed phylogenetic tree of genus *Anoxybacillus* based on 16S rRNA gene exhibited the clustering of TPH1 with *A. rupiensis* DSM 17127 and *A. geothermalis* ATCC BAA-2555 on a single node (Fig. 1A). Further delineation of TPH1 into species level was achieved based on genomic indices i.e., OrthoANI (%), ANIb (%), dDDH (%) and AAI (%), which conferred TPH1 as *A. rupiensis*. Moreover, the codon tree reconstructed using 1000 single copy genes also displayed clustering of TPH1 and *A. rupiensis* DSM 17127 at a single node, whereas *A. geothermalis* ATCC BAA-2555 was distantly positioned (Fig. 1B). Since, the taxonomic position of TPH1 was resolved by genomic features, therefore, the strain was considered as genomospecies of *A. rupiensis*, thus, mentioned as *A. rupiensis* TPH1.

Comparative genomic analysis showed that the genome size of the genus *Anoxybacillus* ranged from 2.6 Mb to 3.6 Mb, with the exception of *A. geothermalis* (6.9 Mb). The G + C content within the genus ranged from 38.9% to 46.8% whereas the total number of CDS within the genus ranged from 2776 in *A. suryakundensis* to 8146 in *A. geothermalis* (Table 1). Codon tree of these *Anoxybacillus* spp. showed two distinct clades i.e. A and B (Fig. 1B). The average genome size of species subsumed in the clade A was found to be 3.47 Mb, which was higher than the species of clade B having average genome size of 2.7 Mb. Consequently, the species of clade A consisted of more CDS and hypothetical proteins than those in clade B.

Further, the pan genome study indicated an open pan genome of genus *Anoxybacillus* and therefore, subsequent addition of new genomes would lead to the inclusion of novel gene families (Fig. 3). This is possibly because *Anoxybacillus* is a relatively new genus and only 19 whole-genome sequences of type species are available. The core genome accounted for 10.62% of the total pan genome and exhibited a gradual declining curve thus, the framework of core genome will not be altered significantly upon the addition of new genomes. Additionally, the unique genes accounted for ~43.5%, where maximum number of unique genes was observed in the genome of *A. geothermalis*. Interestingly, the *Anoxybacillus* spp. subsumed within the clade A consisted of 35.23% unique genes, however the species within clade B harbored much lower percentage of unique genes (8.22%). The presence of more unique genes in the clade A species suggested the variability in habitat and environmental conditions in which they inhabited compared to those in clade B.

COG and KEGG analysis of *Anoxybacillus* spp. revealed that the highest number of core, accessory and unique genes were involved in metabolism related processes, whereas least number of core, accessory and unique genes were associated with cellular processes & signaling and organismal system (Fig. 4).

Furthermore, the genome of *A. rupiensis* TPH1 subsumed 214 and 319 genes involved in carbohydrate and amino acids

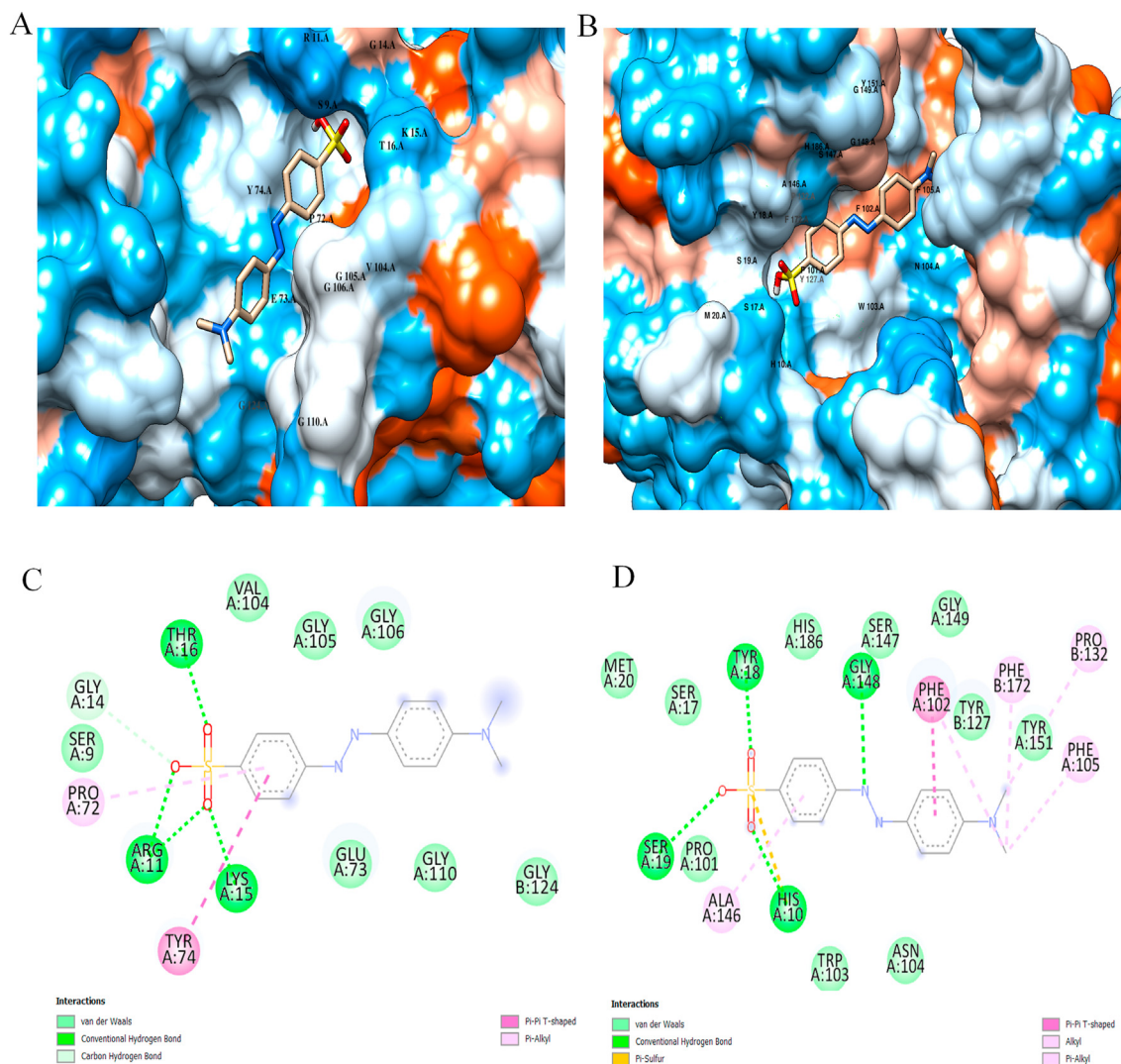


Fig. 6. Molecular docking of azoreductases with MO (A) & (B) The docked complex of AzoRed1-MO and AzoRed2-MO depicting the binding pocket accommodating the substrate along with the interacting residues shown with a single-letter amino acid code and position specifier (C) & (D) The 2-D type interaction between ligand MO with different amino acids in the active site of AzoRed1 and AzoRed2, respectively.

metabolism, respectively (Tables S4 and S7). Among enzyme classes, hydrolases are abundantly harbored in the genome of *A. rupiensis* TPH1 (Fig. 2C). This enzyme class includes proteases, nucleosidases, lipases, phosphatases etc. which are involved in fundamental cellular processes like lipid metabolism, purine nucleotide degradation, programmed cell death etc. [36]. Subsequently, the biochemical assays revealed amylase and caseinase activity in *A. rupiensis* TPH1 (Table S2) which were economically important for their wide range of applications in industries such as textile, paper, detergent etc. [37]. Thus, considering the moderate thermophilic nature of TPH1, these proteases may serve as a potential candidate in several biotechnological industries. The second most abundant class of enzymes in *A. rupiensis* TPH1 was transferases (Fig. 2C). The enzymes subsumed within this class are involved in post translational modifications of polypeptides, kinase activity, sulfhydryl transfer etc. [38] and thus, possibly provide additional stability to the cellular processes in *A. rupiensis* TPH1 at higher temperatures. The third abundant class of enzymes in *A. rupiensis* TPH1 was oxidoreductases which include peroxidases, hydroxylases, reductases and oxygenases [39]. These enzymes seemed to play vital role in ensuring redox homeostasis in the

thermophile through scavenging reactive oxygen and nitrogen species, thus providing antioxidative defense mechanisms and stress mitigation in harsh environments.

The study also reflected the occurrence of a large number of genes coding for various industrially valuable enzymes in the genome of *A. rupiensis* TPH1 (Table S7 A). The majority of the identified enzymes were observed to be involved in the carbohydrate mineralization process whereas few found with lipid metabolism efficiency. The existence of these enzyme coding genes in the *A. rupiensis* TPH1 genome suggested that this bacterium is a prolific source of important enzymes and can be exploited in the industries. The *A. rupiensis* TPH1 was found to contain several enzymes involved in exopolysaccharide production as well as their regulation (Table S7 B).

Moreover, the identification of heavy metal detoxification and azo dye degradation gene in TPH1 conferred its bioremediation properties. Examination of genes responsible for imparting Zn, Cu, Co, Ni, Hg, Fe, Cd and As metal resistance in TPH1 revealed presence of metallochaperones, metalloenzymes, metal-sensing transcriptional regulators and metal transporters (Table S8 A). Genes associated with zinc homeostasis present were transporters,

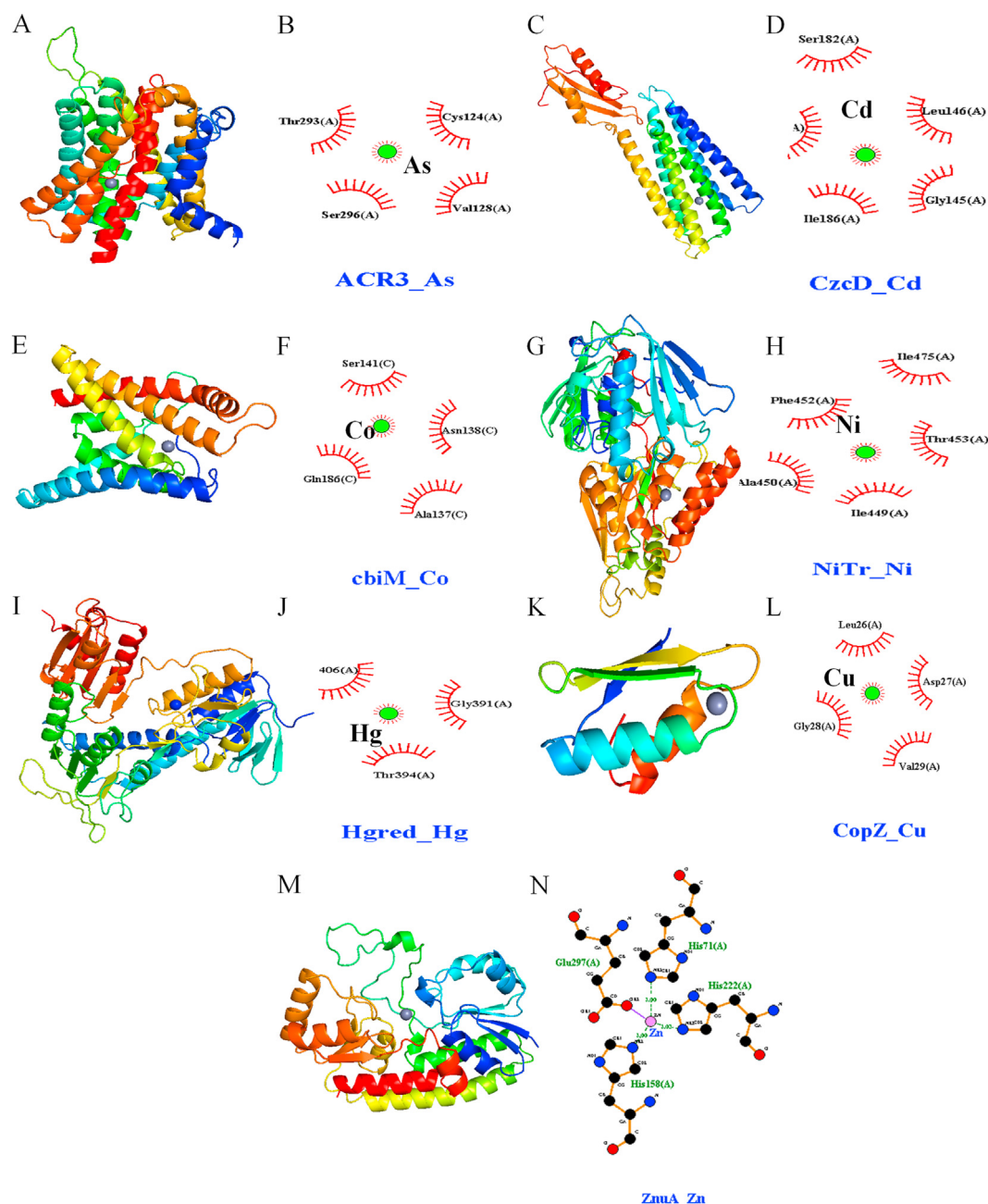


Fig. 7. Molecular docking with interacting residues of metal resistance associated proteins with their respective metals (A) & (B) ACR3_As (C) & (D) CzcD_Cd (E) & (F) cbiM_Co (G) & (H) NiTr_Ni (I) & (J) Hgred_Hg (K) & (L) CopZ_Cu and (M) & (N) ZnuA_Zn.

transcriptional regulators and zinc binding proteins [40] whereas copper homeostasis mediating were cuproenzymes and copper chaperone [41]. Iron homeostasis is regulated at the level of siderophore biosynthesis like bacillibactin and ferrichrome as well as via transporters [42]. Influx and efflux transporters were found in the genome of TPH1 for nickel and cobalt homeostasis [43]. Apart from transporters, reductases and regulation at transcriptional level might be responsible for providing arsenic and mercury tolerance as inferred by the presence of these genes [44,45]. Detoxification against cadmium might be achieved by the presence of transporters and transcriptional regulators. Not any specific gene related to chromium was found in the genome of TPH1. Comparative distributional analysis of selected bioremediating genes within the genus *Anoxybacillus* depicted znuA, copZ and arsC as core genes

whereas genes like czcD, merA, acr3, cbiM, nika and feoB were accessory genes to the genus *Anoxybacillus*. Interestingly, the nika gene was harbored only by the members of clade A in the codon tree (Table S8 B). Subsequently, the molecular docking analysis of these genes with respective metals exhibited lower binding energy conferring stable binding of metals, which further suggests ability of metal sequestration and the involvement of enlisted genes in bioremediation (Fig. 7). Also, growth responses showed survival of TPH1 at toxic concentration of these heavy metals (Fig. S4).

Study on another bioremediating core gene i.e., azoreductase genes was performed. Azoreductases are flavoenzymes, potentially reduces azo bonds (-N=N-) with the help of electron donors via Ping Pong Bi Bi mechanism [46]. Domain analysis depicted the presence of two types of azoreductases i.e., FMN-binding NADPH

Table 1
Comparative genomic features among 19 *Anoxybacillus* spp. including the isolated thermophile *Anoxybacillus rupiensis* TPH1.

Strains	NCBI Taxon ID	Habitat	Country	Genome Status	Contigs	N50	L50	Genome size (Mb)	CDS	HP	GC (%)	tRNA	rRNA	VF	AR	Reference
<i>Anoxybacillus rupiensis</i> TPH1	311460	Hot spring	India	WGS	22	436469	3	3.70	4131	1426	42.3	65	5	2	35	This study
<i>Anoxybacillus rupiensis</i> DSM17127	311460	Hot spring	Bulgaria	WGS	27	3,95,749	4	3.70	4099	1400	42.3	75	7	2	35	[47]
<i>Anoxybacillus amylolyticus</i> DSM15939	294699	Geothermal soil	Antarctica	WGS	3	2834744	1	3.16	3423	1006	43.6	82	22	2	28	[48]
<i>Anoxybacillus ayderensis</i> AB04	265546	Hot spring	Turkey	WGS	74	112260	6	2.83	2959	596	41.8	88	44	3	25	[49]
<i>Anoxybacillus caldiproteolyticus</i> DSM15730	247480	Sewage sludge	Singapore	WGS	30	280,168	5	3.63	3996	1259	40.6	80	6	2	29	[50]
<i>Anoxybacillus calidus</i> DSM25220	575178	Soil near thermal power plant	Turkey	WGS	22	287831	5	3.41	3700	1192	38.9	71	7	2	28	[51]
<i>Anoxybacillus flavithermus</i> DSM2641T	33934	Hot spring	New Zealand	Complete	1	-	-	2.81	2988	627	41.7	83	17	3	28	[52]
<i>Anoxybacillus flavithermus</i> subsp. <i>yunnanensis</i> str.E13	1380408	Hot spring	China	WGS	67	1,03,171	9	2.77	3058	567	41.4	75	11	3	29	[53]
<i>Anoxybacillus geothermophilus</i> ATCC BAA2555	1490051	Mineral deposits in geothermal station	Germany	WGS	333	54,491	35	6.98	8146	2812	46.8	72	5	4	74	[54]
<i>Anoxybacillus gonensis</i> G2	198467	Hot spring	Turkey	WGS	56	1,89,820	5	2.76	2900	574	41.4	75	4	3	27	[55]
<i>Anoxybacillus kamachitensis</i> DSM14988	196180	Hot spring	Russia	WGS	19	2,96,635	4	2.56	2765	550	41.7	64	6	3	28	[56]
<i>Anoxybacillus mongoliensis</i> DSM19169	452565	Hot spring	Mongolia	WGS	35	1,51,781	6	2.63	2830	554	41.4	61	7	3	26	[57]
<i>Anoxybacillus pushtinoensis</i> K1	150248	Manure	Russia	WGS	117	40,724	21	2.63	2952	557	42.1	43	7	3	24	[4]
<i>Anoxybacillus suryakundensis</i> DSM27374	1325335	Hot spring	India	WGS	41	-	-	2.59	2776	477	41.5	61	10	3	24	[58]
<i>Anoxybacillus tengchongensis</i> DSM27374	576944	Hot spring	China	WGS	29	2,04,425	5	2.73	2921	586	41.4	56	5	3	26	[59]
<i>Anoxybacillus tepidamans</i> DSM23211	265948	Geothermal Soil	USA	WGS	54	1,14,465	8	3.38	3707	1057	42.5	73	7	2	24	[60]
<i>Anoxybacillus thermarum</i> AF/04	404937	Thermal mud	Italy	WGS	159	37,018	24	2.74	3260	949	41.9	84	14	3	24	[61]
<i>Anoxybacillus vitaminiphilus</i> CCMCC1.8979	581036	Hot spring	China	WGS	58	1,60,512	6	3.56	3808	1129	40.1	73	8	2	27	[62]
<i>Anoxybacillus voinovskensis</i> DSM17075	230470	Hot spring	Russia	WGS	58	1,06,737	10	3.24	3561	1050	43.94	83	6	2	27	[63]

CDS: Coding DNA sequences; HP: Hypothetical proteins; VF: Virulence factor; AR: Antibiotic resistance gene.

and FMN-binding NADH azoreductase in *Anoxybacillus* spp. (Table S9). The former utilizes NADPH whereas latter utilizes NADH as the electron donor for its activity. Uneven distribution of azoreductases were observed among the genus *Anoxybacillus* since, FMN-binding NADH azoreductase was harbored by the members of both clade A and clade B, while the FMN-binding NADPH azoreductase was distributed only within the members of the clade A (Fig. 1B). The wide distribution of FMN-binding NADH azoreductase enzyme suggests the preference of NADH as an electron donor over NADPH among *Anoxybacillus* spp.

The protein core of theoretical models displayed α/β fold. The FMN-binding NADPH azoreductase (AzoRed1) of *A. rupiensis* TPH1 consisted of 7 β -sheets, among which 5 parallel β -sheets are arranged in 2-1-3-4-5 fashion, and 8 α -helices, whereas FMN-binding NADH azoreductase (AzoRed2) is consists of 7 β -sheets, also comprised of 5 parallel arranged in 2-1-3-6-7 fashion, and 9 α -helices (Fig. 5). Molecular docking studies further corroborated the azo dye degrading ability of AzoRed1 and AzoRed2. It was observed that the binding energy of MO with AzoRed1 and AzoRed2 were -7.2 kcal/mol and -7.7 kcal/mol, respectively, signifying high affinity of MO for these azoreductases. The stability of MO on azoreductases upon docking was probably assured by hydrogen bonds, van der Waal forces, Pi and alkyl interactions between amino acid residues and ligand atoms (Fig. 6). The ability of MO dye decolorization (98%) by TPH1 further corroborated the docking results (Fig. S3).

Conclusively, this comparative genome analysis of genus *Anoxybacillus* not only provided insights into the distribution pattern of genes associated with various cellular processes in *A. rupiensis* TPH1 but also gave evidence of large number of genes encoding industrially valuable enzymes and ability to bioremediate various heavy metals and azo dyes. Therefore, *A. rupiensis* TPH1 represents an excellent platform for the production of industrially important enzymes and for bioremediating toxic pollutants like heavy metals and azo dyes.

4.1. Strain deposition and genome accession number

The strain was deposited in National Agriculturally Important Microbial Culture Collection (NAIMCC), ICAR-National Bureau of Agriculturally Important Microorganisms (NBAIM) with accession number NAIMCC-SB-0052. The *Anoxybacillus rupiensis* TPH1 is submitted under BioProject PRJNA738657 and whole-genome accession number is JAHSSG000000000.

Declarations of competing interest

All the authors declare no major conflict of interest.

Authors' contributions

AM and SSS conceptualized the study. AM, SK, SC and TPJ performed the experiments. AM, SK, TPJ, SC and SSS analysed the experimental data. AM, SB, and AKM analysed whole genome sequence. AM, SC, SB, TPJ and SK performed the comparative genomic studies. AM, SB and SC drafted the original article. SC, AKM and SSS reviewed and constructed the final manuscript. All authors contributed to the article and approved the submitted version.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.resmic.2023.104027>.

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